

SEMINAR REPORT

Submitted in partial fulfillment of the requirements for the award of

the degree of

***M.Sc. Computer Science with Specialization
in Artificial Intelligence***

2024-2026

By

John Abraham Chandy (
24030194)

DEPARTMENT OF COMPUTER APPLICATIONS



**COCHIN UNIVERSITY OF SCIENCE
AND TECHNOLOGY, COCHIN-22**

NOVEMBER 2025

**DEPARTMENT OF COMPUTER APPLICATIONS
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY COCHIN-22**



CERTIFICATE

This is to certify that the seminar report on the topic “**Transformative Role of Deep Learning in Video Games** ” submitted in partial fulfillment of the requirements for the award of the Degree of **M.Sc. Computer Science with a Specialization in Artificial Intelligence** is a record of bona fide work done by **John Abraham Chandy** during the period from **July 2025 to November 2025** of their study in the Department of Computer Applications at Cochin University of Science and Technology.

DECLARATION

I hereby declare that the seminar report on the topic “**Transformative Role of Deep Learning in Video Games**” submitted in partial fulfillment of the requirements for the award of the Degree of **M.Sc. Computer Science with a Specialization in Artificial Intelligence** is a record of bona fide work done by me during the Third Semester of my course MSc Computer Science with specialization in Artificial Intelligence, under the supervision and guidance of **Dr. Rafidha Rehiman K. A.** in the Department of Computer Applications, Cochin University of Science and Technology, Cochin-22.

Place: CUSAT
Date: 10/11/20205

Signature
John Abraham Chandy (24030194)

ACKNOWLEDGEMENT

I owe it to the people who helped me with this report. My sincere thanks to my advisor, Dr. Rafidha Rehiman K. A., for her steady support, sharp feedback, and encouragement from start to finish. Thanks to all the faculty and staff in my department for creating an open, helpful space and giving me everything I needed for my research. I'm also grateful to my friends and classmates for their support. Most of all, I want to thank my family. Their belief in me and constant encouragement made all the difference.

Place: CUSAT
Date: 10/11/20205

Signature
John Abraham Chandy (24030194)

TABLE OF CONTENTS

Section	Page No.
Abstract	5
1.Introduction	6
2.1 Deep Learning Overview	7
2.1.1 Reinforcement Learning	7
2.1.2 Procedural Content Generation	7
2.2 Advanced Architectures	8
2.2.1 Transformer Models	8
2.2.2 Multi-Agent Learning	8
2.3 Specialized Methods	9
2.3.1 SimBa Real-Time Behavior	9
2.3.2 Residual-MPPI Control	10
3. Conclusion	11
4. Future Directions	11
5. References	12
Appendix	12
Plagiarism Report	13

ABSTRACT

Deep learning is shaking up the video game world. It's not just about better graphics or smarter characters—these technologies are changing how games feel and respond. Think about NPCs that actually learn, game levels that build themselves, and play experiences that shift to match your style. This report dives into those breakthroughs, from transformer-based reinforcement learning to multi-agent systems and smarter control frameworks. The main takeaway: deep learning is pushing games to be more complex, more believable, and way more engaging.

Terminology Used

Deep Learning – A branch of machine learning that uses neural networks to recognize patterns and solve complicated problems.

Reinforcement Learning – Training agents to make good decisions by rewarding the right moves in a simulated environment.

Procedural Content Generation – Using algorithms to automatically make game content like maps, levels, or items.

Adaptive Gameplay Personalization – Tweaking the game's behavior and difficulty on the fly, based on how the player acts.

1. INTRODUCTION

Deep learning has fundamentally transformed the video game industry, ushering in a new era of intelligent, adaptive, and immersive gaming experiences. Traditionally, game design relied heavily on pre-scripted behaviors for non-player characters (NPCs) and manually crafted levels and assets. This approach, while effective, often led to predictable gameplay and limited creative scope for developers. However, with the advent of deep learning, NPCs are now capable of autonomous decision-making, learning from player interactions, and adapting in real time, which results in richer, more dynamic gameplay.

In addition to enhancing the behavior of game characters, deep learning techniques have also revolutionized other aspects of game development. Procedural content generation powered by AI can create vast, unique game worlds and assets without the need for extensive manual input, significantly reducing development time and enabling unprecedented creativity. AI-driven storytelling further enriches the narrative experience by dynamically tailoring story arcs based on player choices, thus personalizing every playthrough.

This report delves into several groundbreaking approaches at the intersection of AI and gaming which illustrate these advancements, including Transformer-Based Reinforcement Learning that improves decision-making processes, Multi-Agent Deep Reinforcement Learning (MADRL) for complex interactions among multiple entities, and Procedural Content Generation coupled with AI-Driven Storytelling for dynamic environments and narratives. Real-time behavioral modeling using SimBa demonstrates how NPCs can act plausibly in fast-paced scenarios, while Adaptive Control methods like Residual-MPPI illustrate how AI can maintain optimal performance in uncertain and changing game conditions.

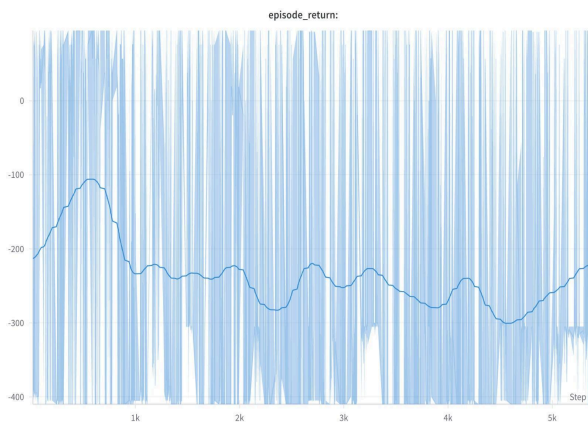
Through these topics, the report aims to showcase how deep learning is not only evolving the intelligence and autonomy of game elements but also empowering developers to create more engaging, unpredictable, and lifelike virtual worlds for players. These advances mark a significant shift from traditional static game design to dynamic ecosystems where both NPCs and game environments continuously learn, adapt, and respond to the player's actions.

2.1 DEEP LEARNING OVERVIEW

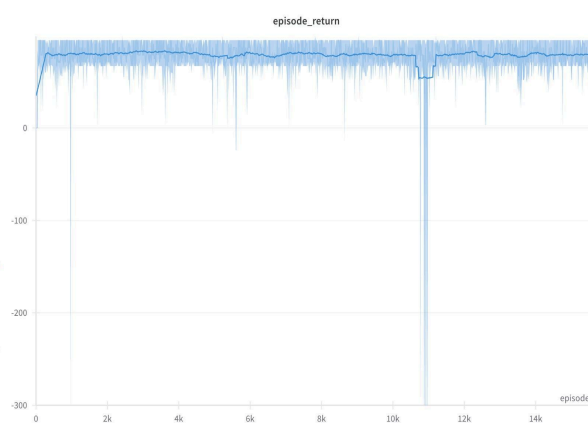
Deep learning is a type of machine learning that relies on neural networks to spot patterns and learn from data. In gaming, it's a game-changer. For example, with reinforcement learning, agents pick up the best strategies through rewards and feedback, leading to smarter, more responsive gameplay.

2.1.1 Reinforcement Learning (RL)

Reinforcement learning teaches game agents how to make decisions by throwing them into the game world and letting them figure things out, always chasing higher rewards. Platforms like VizDoom push these agents to learn skills you'd expect in a first-person shooter: moving around, fighting, coming up with strategies—actually playing the game, not just following scripts. Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO) have stepped up the game here, making AI more adaptable and complex.



Results - DT vs PPO [Ref: 1]



Results - DQN+DRQN vs DTQN [Ref: 1]

2.1.2 Procedural Content Generation (PCG)

Procedural Content Generation, or PCG, lets AI whip up game content—levels, landscapes, even storylines—without endless manual design. It's all powered by algorithms like GANs or language models such as GPT-2. Developers save tons of time, and players keep getting new, unpredictable experiences every time they play. It's a boost for replay value and keeps things fresh.

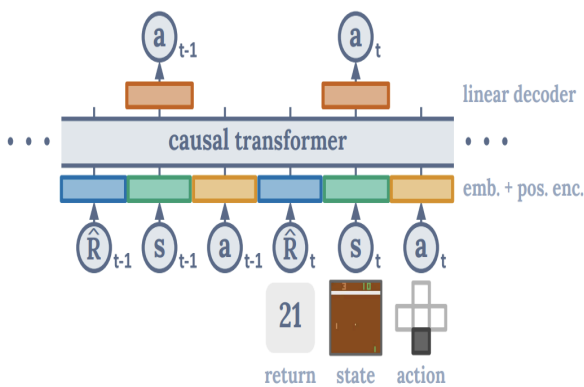
But you can not just let AI run wild. If you care about fairness and creativity, you need to make sure PCG follows ethical guidelines. Content should be inclusive, free from bias, and respect players' privacy. Automation makes things faster and more creative, but real people need to keep an eye on the process. That's how you keep things original and in line with ethical standards. When you blend AI's speed with

thoughtful human oversight, you end up with game worlds that are dynamic, engaging, and fair—and keep surprising players.

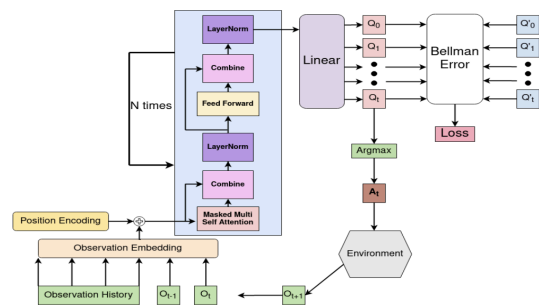
2.2 ADVANCED ARCHITECTURES

2.2.1 Transformer Models in Games

Transformers are showing up in game AI now, especially for handling problems like limited information and planning far ahead—things that trips up other models. Decision Transformers, for example, treat reinforcement learning as a sequence modeling problem, so they can learn from offline data. They work well in some games, especially where remembering long sequences matters, but they are not a magic fix for every scenario.



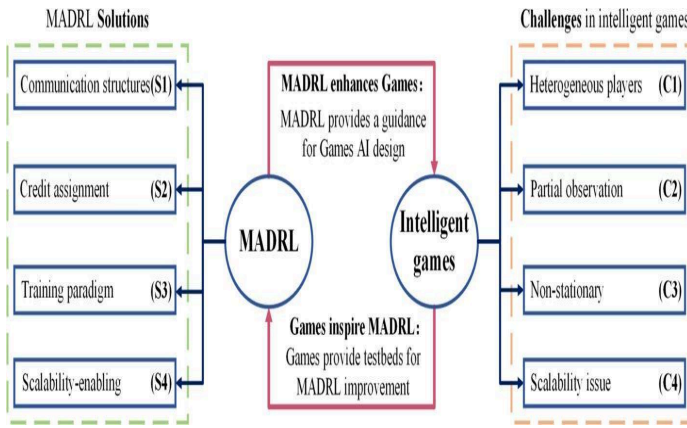
DTQN Architecture [Ref: 1]



Decision Transformer [Ref: 1]

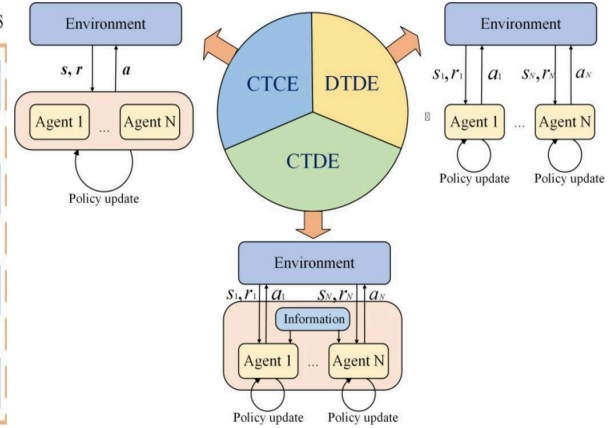
2.2.2 Multi-Agent Deep Reinforcement Learning (MADRL)

Multi-Agent Deep RL takes things up a notch by throwing several agents into the same environment. This is key for games where teamwork or competition matters. These agents need to deal with stuff like different skills, limited ways to talk to each other, and environments that keep changing.



MADRL Architecture [Ref: 2]

[Ref: 2]



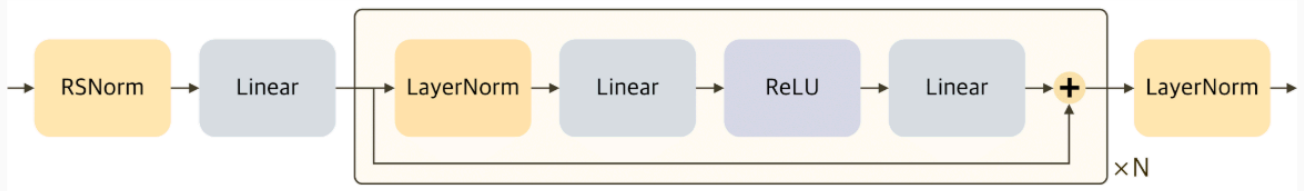
Training & Execution Paradigms in MARL

2.3 SPECIALIZED METHODS

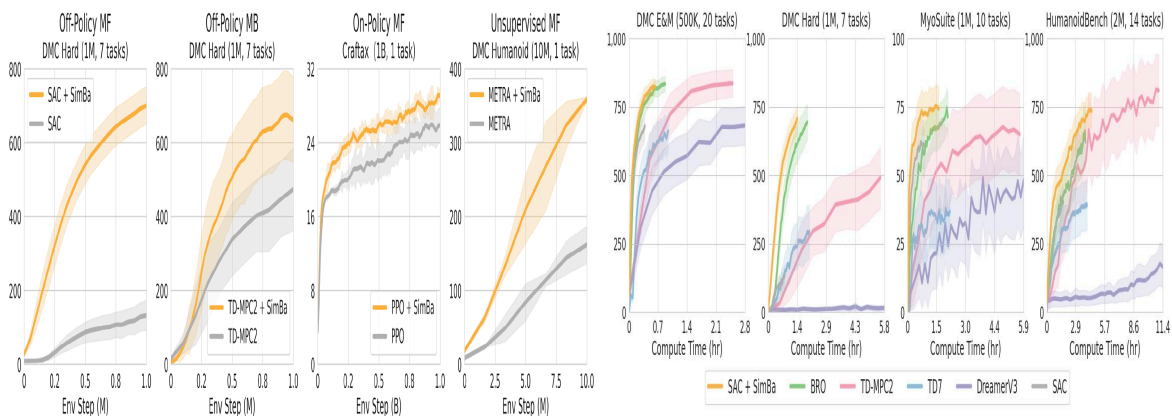
2.3.1 SimBa for Real-Time Behavior Adaptation

SimBa brings a simplicity bias to reinforcement learning, making training more stable and helping agents generalize better in real-time games. It helps NPCs learn efficiently and adapt on the fly, which makes their behavior more believable and keeps players immersed.

SimBa Architecture

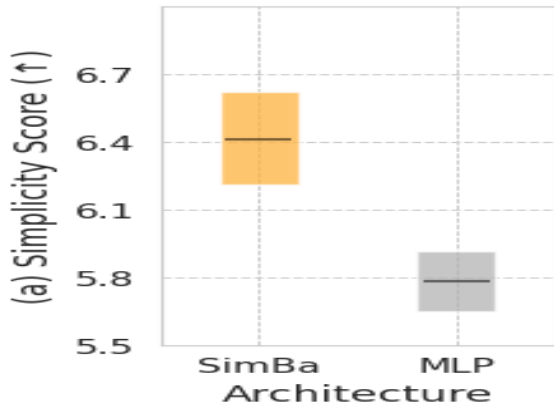


SimBa comprises three components: Running Statistics Normalization, Residual Feedforward Blocks, and Post-Layer Normalization. These components lower the network's functional complexity, enhancing generalization for highly overparameterized configurations.

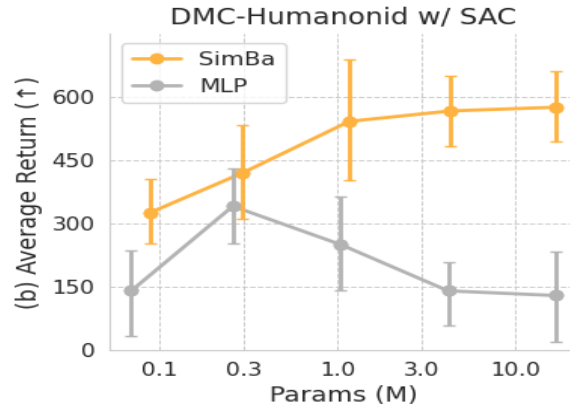


SimBa Scaling Superiority

SimBa Efficiency



Simplicity Score Comparison

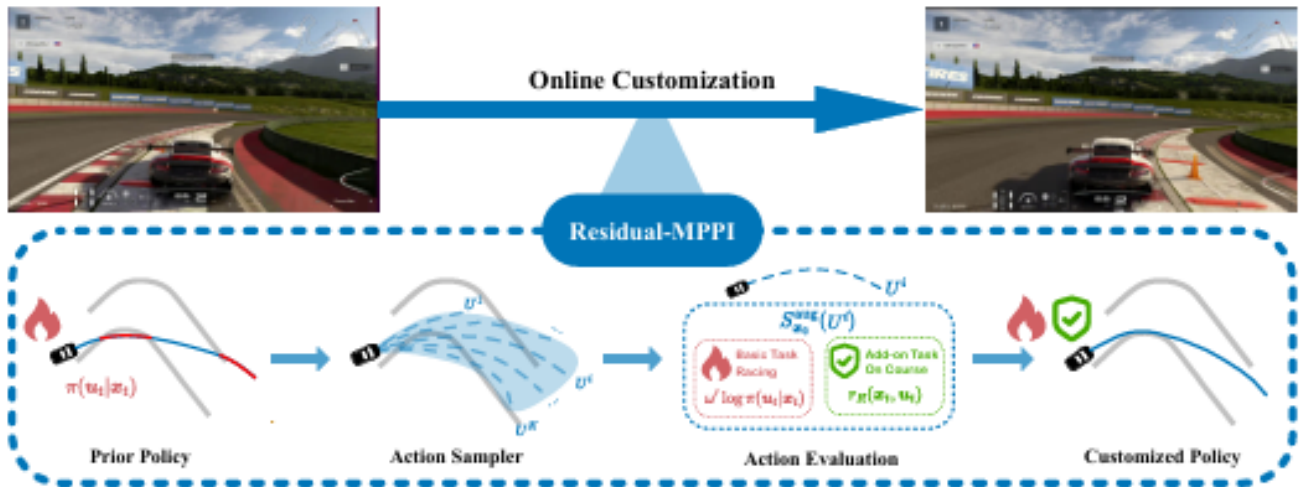


SimBa Outperforms MLP

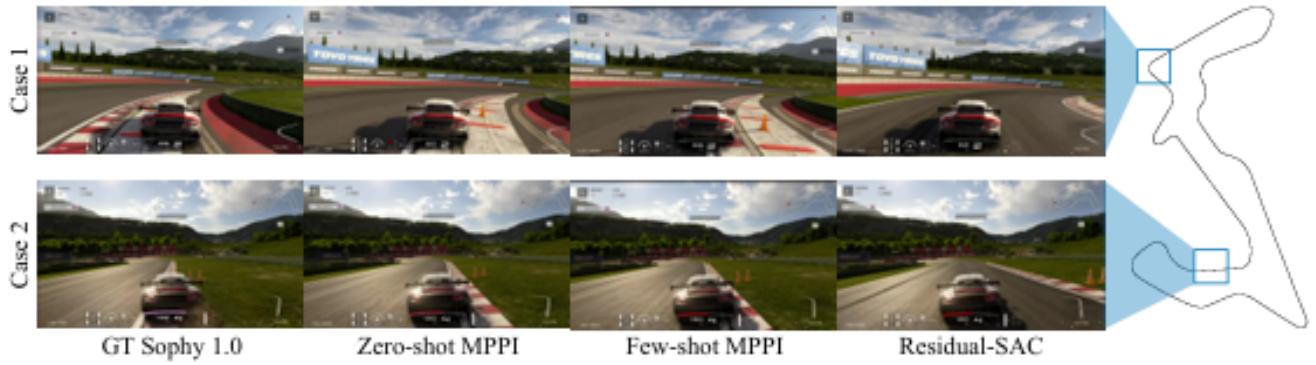
[Ref: 4 include above diagrams and directly copied materials with appropriate citations.]

2.3.2 Residual-MPPI Adaptive Control

Residual Model Predictive Path Integral (MPPI) control gives game characters and vehicles a way to move that's both smooth and adaptable. By mixing classic MPPI with residual learning, this method helps AI react to unpredictable situations, keeping movement responsive and always in tune with the game's goals.



[Residual-MPPI Adaptive Control Framework { Ref: 5 }]



[Cornering Strategy Analysis with Residual-MPPI { Ref: 5 }]



[Trajectory Guidance Comparison with Residual-MPPI { Ref: 5 }]

3. CONCLUSION

<u>Applicables and Benefits</u>		
Technique	Application	Benefits
Transformer-Based Reinforcement Learning	Training complex game AI agents (e.g., <i>StarCraft II</i>)	Handles long-term planning, improves strategic decision-making, learns complex patterns
Multi-Agent Deep Reinforcement Learning (MADRL)	Team-based AI coordination in cooperative games	Enables collaboration, realistic group behavior, emergent strategies
Procedural Content & AI-Driven Narrative	Dynamic storylines and levels (e.g., <i>AI Dungeon</i>)	Increases replayability, personalized experiences, reduces manual content creation
SimBa for Real-Time	Adaptive NPC behavior	NPCs respond realistically to

<u>Applicables and Benefits</u>		
Behavior		player actions, improves immersion
Residual-MPPI for Adaptive Control	Real-time vehicle or robot control in simulation games	Smooth and adaptive motion, robust performance in unpredictable environments

Deep learning has completely changed how video games get made. Games now feel smarter, more responsive, and way more immersive. Look at transformer-based RL, multi-agent teamwork, or new adaptive controllers—they all make gameplay feel more realistic and layered. With these tools, developers can build games that grab you, surprise you, and adjust to the way you play.

4. FUTURE DIRECTIONS

Looking ahead, research is zeroing in on making AI work better across all sorts of games, not just one genre. There's also a push to make AI decisions easier to understand and more transparent, so players know what's going on behind the curtain. Large language models are starting to spin richer, more involved game narratives. Plus, faster and more efficient training methods, along with scalable ways to work together, are set to shake up game AI even more.

5. REFERENCE

1. Batth, S., Sethi, H., Shariff, Z., Shi, J., Patel, R. (2025). Do We Need Transformers to Play FPS Video Games? arXiv preprint arXiv:2504.17891.
2. Wang, T., et al. (2025). Intelligent games meeting with multi-agent deep reinforcement learning: A comprehensive review. Artificial Intelligence Review.
3. Li, X., et al. (2025). The Exploring of AI Applications in Game Development. Applied and Computational Engineering. ResearchGate.
4. Sony AI. (2025). SimBa Simplicity Bias for Real-Time Game Agents. Presented at ICLR 2025.

5. Sony AI. (2025). Residual-MPPI Hybrid Control for Personalized Game AI. Presented at ICLR 2025.

[References have all diagrams and directly copied materials with appropriate citations.]

Appendix

Significance of Topic and Novelty

This report digs into how deep learning and video games collide, showing how AI doesn't just make games more fun but also raises the bar for what smart game agents can do. What stands out right now is the use of transformer models and multi-agent setups—these let game AIs pull off trickier strategies and scale up to handle more complex scenarios. New methods like SimBa and Residual-MPPI are pushing the envelope for real-time adaptation, letting NPCs and in-game agents adjust on the fly. These breakthroughs are not just academic—they matter for game studios and players alike. As games keep evolving, these AI advances shape both the science and the business of interactive entertainment.

Plagiarism Report

Plagiarism Detection by Paperrater

Original Work

Originality: 93%

This paper appears to be original

AI Detection by NoteGPT.io



NoteGPT Certification

